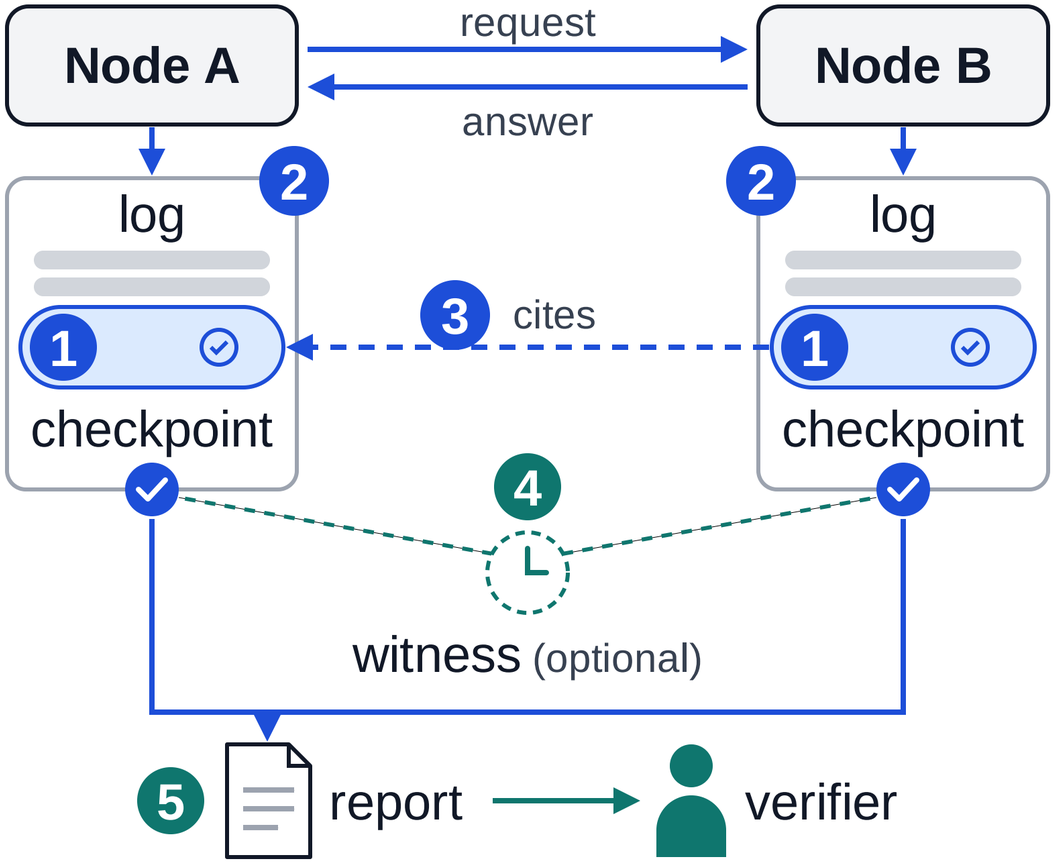
Mesh-LLM Evidence Layer: What Trust between Strangers Looks Like

What a node says it served, and its history of what it has done (targeting mesh-llm v0.79)



1. **Capsule:** a signed record of one action. Digests, never the prompt or answer.
2. **Log + checkpoint:** append-only and signed, with no central server.
3. **Bilateral:** both sides record the same exchange, so any gap or mismatch shows.
4. **Witness:** an outside timestamp. It shows when, not what.
5. **Third-party report:** anyone checks it offline, with no account.

“The mesh needs no central authority to trust: each node keeps its own receipt and checks the other’s.”

What each check shows, and what it doesn’t

Question — Check	Shows	Does not show
Who sent this, and was it changed? — **Signed**	who signed; unchanged since	that the content is true
Was it there at the time? — **Checkpointed**	in the log by time T, in order	that everything was logged
Did we both see the same exchange? — **Both sides match**	same request and answer, cited both ways	which side is right
Was the claimed model served? — **Second node + referee**	a signed ruling on one client-marked pair	every request; the referee is off by default today
What actually ran? — **Hardware attestation (next)**	a measured runtime	more than the quote covers

What keeps everyone honest

- each side’s own signed log
- the other side’s copy
- an outside witness
- a referee neither side picked

→ builds a history per peer
(counts, never a score)

Go deeper

agentactioncapsule.org
SCITT, RFC 9943

datatracker.ietf.org/doc/search?name=draft-mih
(AAC, checkpointed log, bilateral, evidence layer; individual Internet-Drafts)
github.com/Mesh-LLM/capsules (evidence plugin)

“No reputation score anywhere. You get a map of what was checked, and the gaps are named.”